

Public Safety Technology in Chicago: Mapping ALPR Cameras and Community Hardship

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Streamlit dashboard: <https://alpr-surveillance-dashboard-axawcpzaowmvest455ladk.streamlit.app/>

GitHub repository: <https://github.com/kikoli0306-source/alpr-surveillance-dashboard>

Research Question and Data

This project examines where recorded automated license plate reader (ALPR) cameras appear in Chicago and how their distribution relates to community-level socioeconomic hardship. The goal is not to prove causation, but to make spatial patterns of public safety technology visible.

I used one primary dataset and two supporting datasets. The ALPR camera locations come from DeFlock's public camera map: <https://maps.deflock.org>. The Chicago community area boundaries come from the Chicago Data Portal: <https://data.cityofchicago.org/Facilities-Geographic-Boundaries/Boundaries-Community-Areas-Map/cauq-8yn6>. The socioeconomic indicators also come from the Chicago Data Portal: <https://data.cityofchicago.org/Health-Human-Services/Census-Data-Selected-socioeconomic-indicators-in-C/kn9c-c2s2>. These indicators include hardship index, poverty rate, unemployment rate, and per capita income.

I converted the camera coordinates into point data, spatially joined them to Chicago's 77 community area polygons, and aggregated camera counts by community area. The dashboard uses 595 recorded ALPR camera records, 594 of which were successfully matched to community areas.

Visualization Design

I chose the visualizations to move from raw locations to community-level interpretation. The point map shows the original camera locations before aggregation. The side-by-side maps compare recorded camera concentration with selected socioeconomic indicators without implying causation. The quadrant scatter plot separates communities by median hardship and median camera count, making it easier to see where above-median hardship overlaps with above-median camera presence. I also used a proportional symbol map because choropleth maps can overemphasize large areas; circle size represents camera count more directly. The operator chart shows that ALPR records include public, private, institutional, and unknown operators.

Key Findings

First, recorded ALPR cameras are spatially uneven. They are not evenly scattered across Chicago, but cluster along corridors, central neighborhoods, and selected urban areas.

Second, camera-heavy communities are socially mixed. Near North Side, West Town, Lincoln Park, Near West Side, and the Loop have high recorded camera counts, but high-hardship communities such as Austin, Fuller Park, and West Garfield Park also appear in the upper group. This suggests that high camera counts are not limited to one type of community.

Third, hardship is relevant context, but it does not explain camera distribution by itself. The scatter plot shows overlap between above-median hardship and above-median camera count in some communities, but not a simple positive relationship overall. Camera placement may also reflect road networks, traffic flow, commercial activity, institutional locations, and reporting density.

The operator field also requires caution. Of 595 camera records, 319 have unknown operator information and 276 have known labels. Unknown does not mean unmonitored; it means the public record does not include operator metadata.

Limitations and Outlook

This dashboard maps recorded camera presence, not a complete official inventory or how captured data is stored, shared, or used. Raw counts are not normalized by population, road length, traffic volume, or commercial activity. The socioeconomic indicators may also come from a different time period than the camera data. For these reasons, the project shows spatial patterns and associations, not causal relationships.

With more resources, I would add road network data, traffic volume, police district boundaries, and camera installation dates. I would also calculate camera rates per road mile or per traffic volume and add a street-level inspection mode for viewing nearby ALPR cameras within a selected radius.

AI Use Statement

I developed the research question, visualization plan, and interpretation myself. I used Claude for coding support while building the Streamlit dashboard, especially for debugging, Plotly implementation, layout adjustments, and deployment troubleshooting. I reviewed and revised the final dashboard and written analysis before submission.